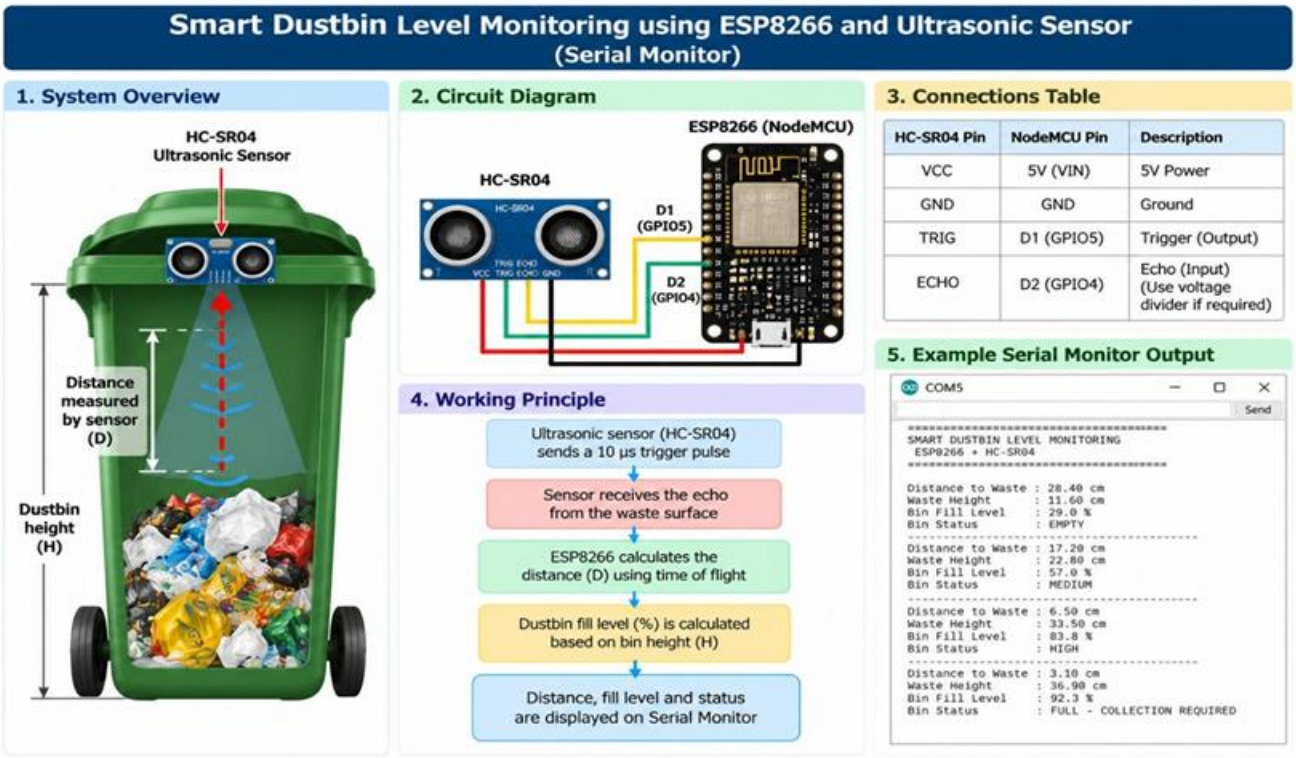
ASSIGNMENT – 8

BATCH 8

Aim: Smart Dustbin Level Monitoring experiment, the ESP8266 can measure the distance from the HC-SR04 mounted at the top of the dustbin and calculate the **waste-fill percentage**. The result can be displayed in the **Serial Monitor**.



How the calculation works:

If your dustbin has an internal height of 40 cm:

1. Waste Height

Waste Height = Bin Height − Distance to Waste

where the distance is the measured distance from the ultrasonic sensor to the waste surface.

2. Bin Fill Percentage

Fill Level (%) = [(Bin Height − Distance) / Bin Height] × 100

Example:

Bin height = 40 cm

Sensor-to-waste distance = 10 cm

Fill Level = [(40 − 10) / 40] × 100 = 75%

Therefore, the dustbin is 75% full and can be classified as HIGH fill level.

Without Blynk

Code:

```
#define SOIL_PIN Ao

void setup() {
  Serial.begin(9600);
}

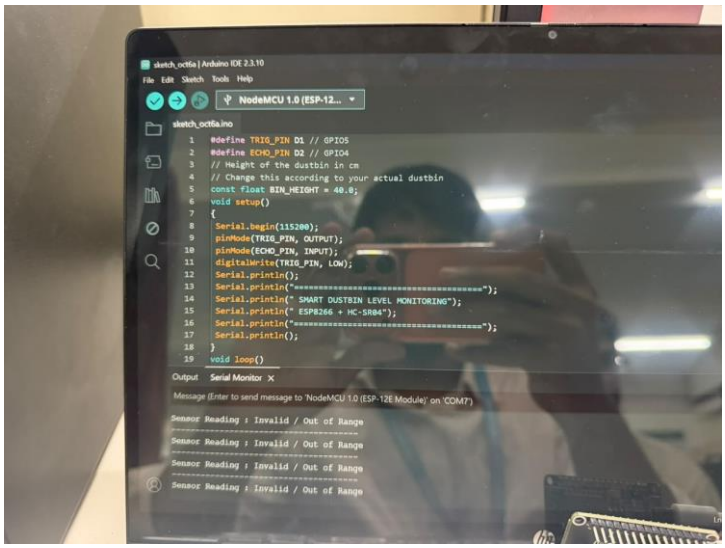
void loop() {
  int moistureValue = analogRead(SOIL_PIN);

  Serial.print("Moisture: ");
  Serial.println(moistureValue);

  if (moistureValue > 700) {
    Serial.println("DRY");
  }
  else if (moistureValue > 400) {
    Serial.println("MOIST");
  }
  else {
    Serial.println("WET");
  }

  delay(1000);
}
```

Output:



With Blynk

Code:

```
#define BLYNK_TEMPLATE_ID "TMPL3iFKyzQSg"
#define BLYNK_TEMPLATE_NAME "Ultrasonic Dustbin"
#define BLYNK_AUTH_TOKEN "m7gQ2ygYpzWi5hvskmlQ-6lYQYtSahiQ"

#include <ESP8266WiFi.h>
#include <BlynkSimpleEsp8266.h>

char ssid[] = "Not for Sale";
char pass[] = "panipuri";

#define TRIG_PIN D5
#define ECHO_PIN D6

BlynkTimer timer;

void sendDistance()
{
  digitalWrite(TRIG_PIN, LOW);
  delayMicroseconds(2);

  digitalWrite(TRIG_PIN, HIGH);
  delayMicroseconds(10);

  digitalWrite(TRIG_PIN, LOW);

  long duration = pulseIn(ECHO_PIN, HIGH, 30000);

  if (duration == 0)
  {
    Serial.println("No echo received");
    Blynk.virtualWrite(V0, 0);
    return;
  }

  float distance = duration * 0.0343 / 2.0;

  Serial.print("Distance: ");
  Serial.print(distance);
  Serial.println(" cm");

  Blynk.virtualWrite(V0, distance);
}

void setup()
{
  Serial.begin(115200);

  pinMode(TRIG_PIN, OUTPUT);
  pinMode(ECHO_PIN, INPUT);

  digitalWrite(TRIG_PIN, LOW);
```

```
    Blynk.begin(BLYNK_AUTH_TOKEN, ssid, pass);

    timer.setInterval(1000L, sendDistance);
}

void loop()
{
    Blynk.run();
    timer.run();
}
```

Output:

